



Applied Machine Learning in Practice: Insights from Real-World Predictive Analytics Case Studies

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How do companies make decisions?

NETFLIX

How do you optimize
what movies to
recommend?

 **NBA**

How to select a new
prospect for your NBA
basketball team?

 **BARCLAYS**

How to decide if a
transaction is fraud or
not?



Well, just look at the data.

Data Driven Decision Making



The process of using analyzed and processed data to guide business decisions





Decisions Based on Predictions



A specific type of decisions:



Make a predictions based on
past data



Make a decision for
the future



Evaluate decision based on
realization of event

Decisions Based on Predictions.



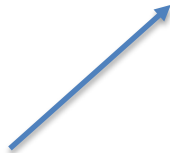
Predictive analytics is the process of **using historical data to predict future outcomes** using data analytics, machine learning, and statistical models

We want to predict the future in order to act now.

$$\text{best action} = \underset{a}{\operatorname{argmin}} \mathbb{E}_{\mathbf{P}}[\text{cost}(a, z_{i,T+1}, z_{i,T+2}, \dots, z_{i,T+h})]$$



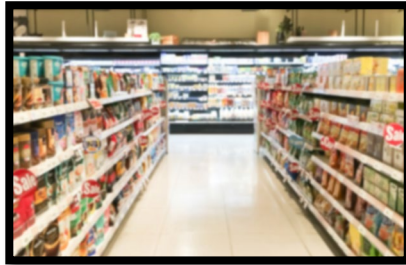
How to obtain $z_{i,T+1} \dots + z_{i,T+h}$



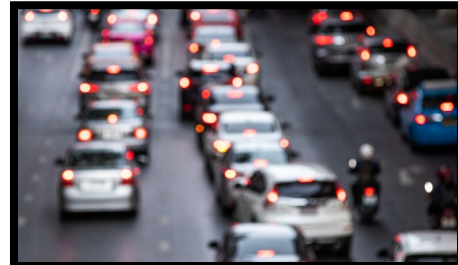
An Explosion of Predictive Analytics Applications



Electricity load



Retail demand



Traffic

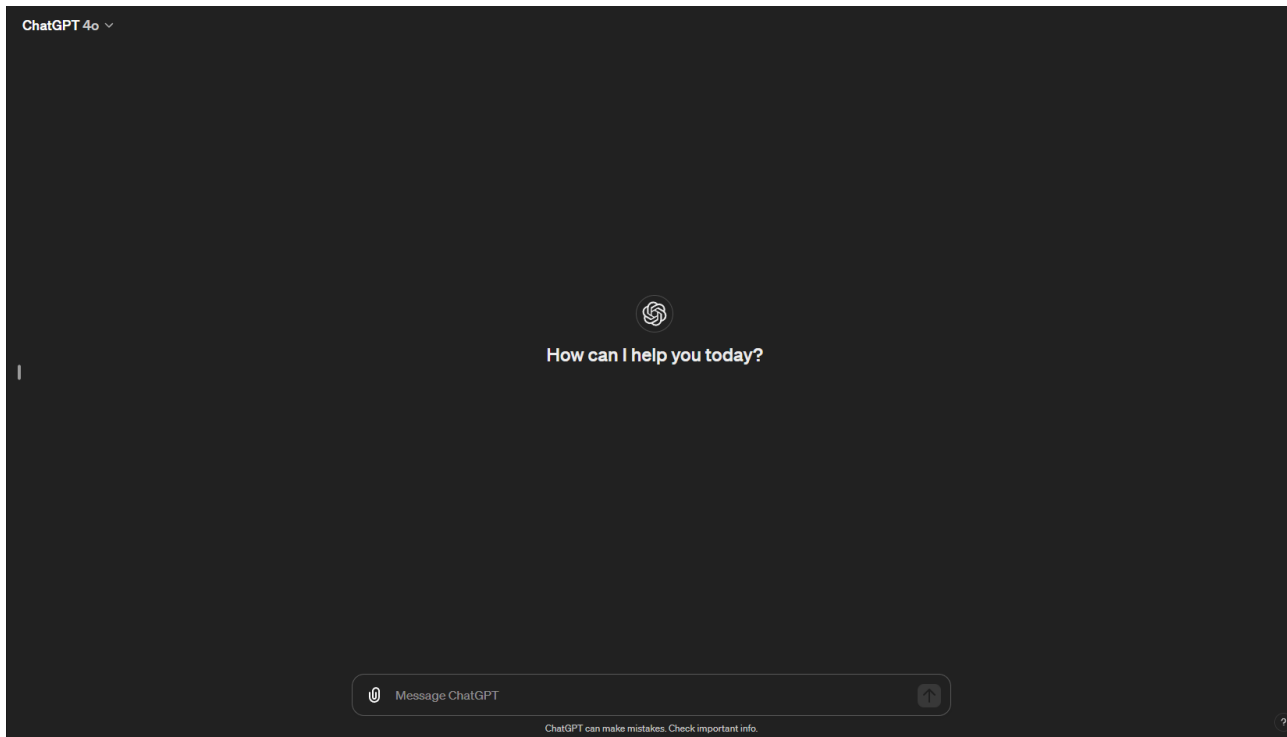


Econometrics

What changed?

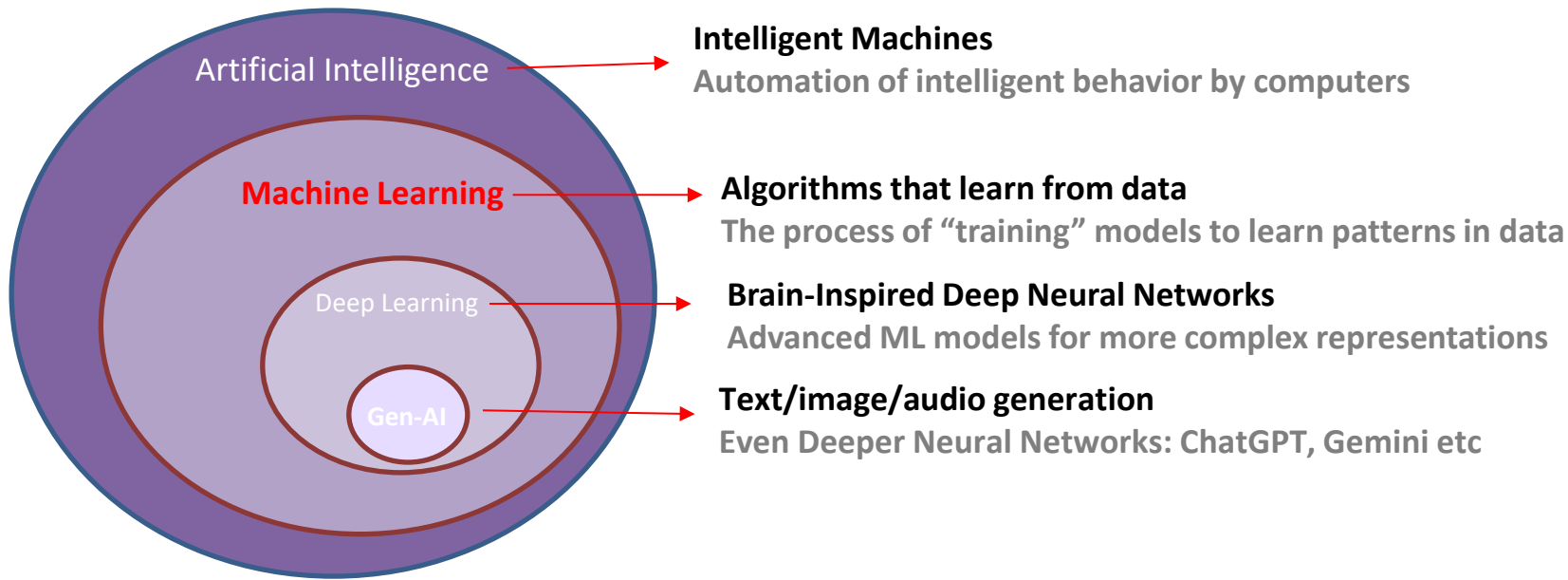
- 1) Better (and more) data
- 2) Better (and more) questions
- 3) Better algorithms. Yes, AI.

What Most People Think About AI?





What is AI?



Machine Learning is much more than modern LLMs

Outline of the Talk:



1. What is Machine Learning (ML) in the context of Predictive Analytics
2. How to apply ML in practice (with examples from real world applications)
 - How to scope?
 - How to collect your data?
 - How to build your solution?
 - How to maintain and evaluate it?
3. What about GenAI and LLMs for Predictive Analytics?



SECTION 1

What is Machine Learning in the Context of Predictive Analytics?

To Start With. What is Machine Learning?

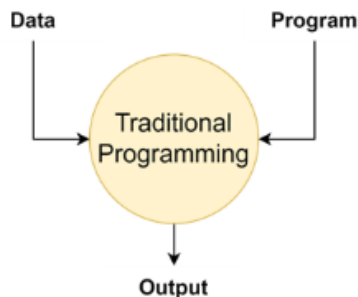
“Field of study that gives computers the ability to learn without being explicitly programmed”

Arthur Samuel (1959)



“A subfield of AI that contains the application and science of algorithms that make sense of data”

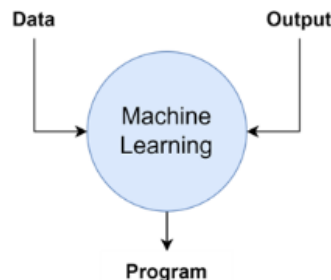
“Machine Learning with Pytorch and Scikit-Learn”



Traditional Programming



DeepBlue 1997

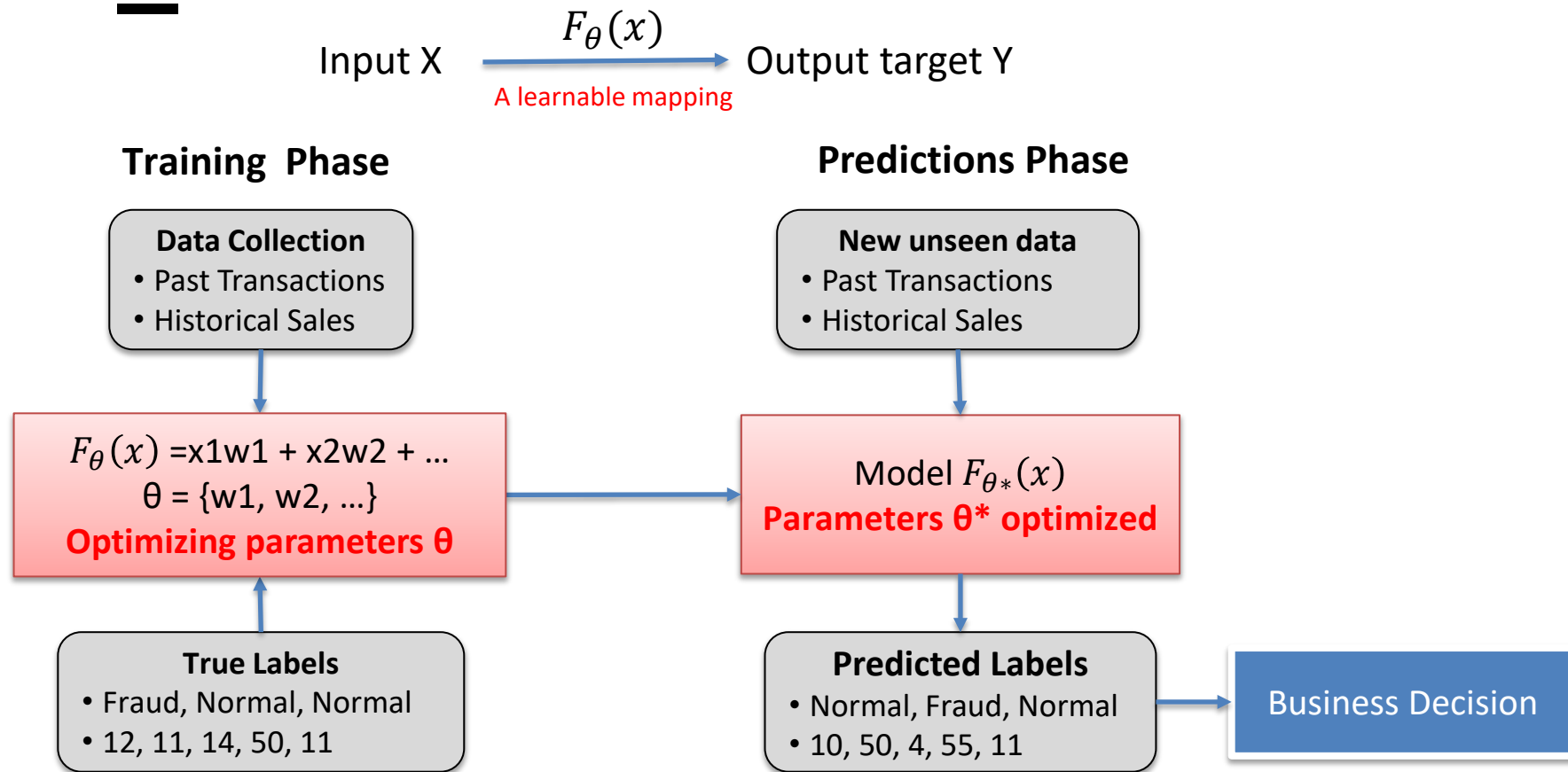


Machine Learning



Alpha Zero 2018

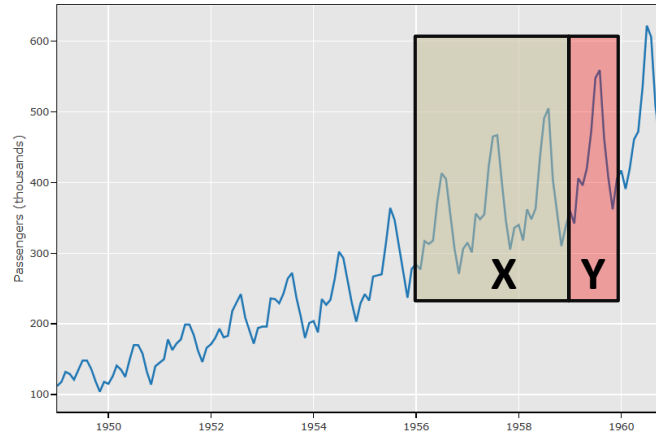
Lets formalize this a bit.



Machine Learning in Predictive Analytics



Past total Passengers X $\xrightarrow{F_{\theta}(x)}$ Future total Passengers Y



This is a time series!

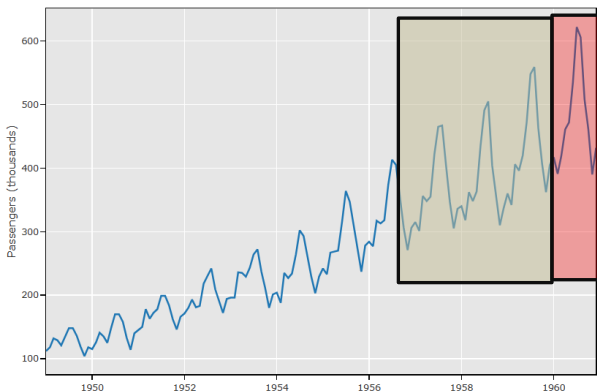
Task:

Predict the total monthly Air Passengers for a full year

Going from X to Y

Past total Passengers X $\xrightarrow{F_{\theta}(x)}$ Future total Passengers Y

Step 1 Tabularize



	Features (X) - Lagged Values				Target (y)
	$X_1(t-4)$	$X_2(t-3)$	$X_3(t-2)$	$X_4(t-1)$	$y(t+1)$
Jun 1949	112	118	132	129	135
Jul 1949	118	132	129	121	148
Aug 1949	132	129	121	135	148
Sep 1949	129	121	135	148	136
Oct 1949	121	135	148	148	119
Nov 1949	135	148	148	136	104

Task:

Predict the total monthly Air Passengers for a full year

Step 2

Pick a Model

$$F_{\theta}(x) = x_1 w_1 + x_2 w_2 + x_3 w_3 + x_4 w_4$$

Step 3

Make a prediction

$$\hat{y}_1 = f(\text{June 1949})$$

Step 4

Calculate a Loss function

$$L = \frac{1}{2m} \sum_{i=1}^m \left(\underset{\text{error}}{\hat{y}^{(i)} - y^{(i)}} \right)^2$$

Step 5

Update weights using gradient descent



Wow Machine Learning is easy, right?

Machine Learning is easy, right?



1. Collect data
2. Train model
3. Use your model for business decisions

Machine Learning is easy, right?



1. Collect data
2. Train model
3. Use your model for business decisions
- 4.



Machine Learning is easy, right?



1. Choose a metric to optimize
2. Collect data
3. Train model
4. Realize many labels are wrong -> relabel data
5. Train model
6. Model performs poorly on one class -> collect more data for that class
7. Train model
8. Model performs poorly on most recent data -> collect more recent data
9. Train model
10. Deploy model
11. Dream about \$\$\$
12. Wake up at 2am to complaints that model biases against one group -> revert to older version
13. Get more data, train more, do more testing
14. Deploy model
15. Pray
16. Model performs well but revenue decreasing
17. Cry
18. Choose a different metric
19. Start over

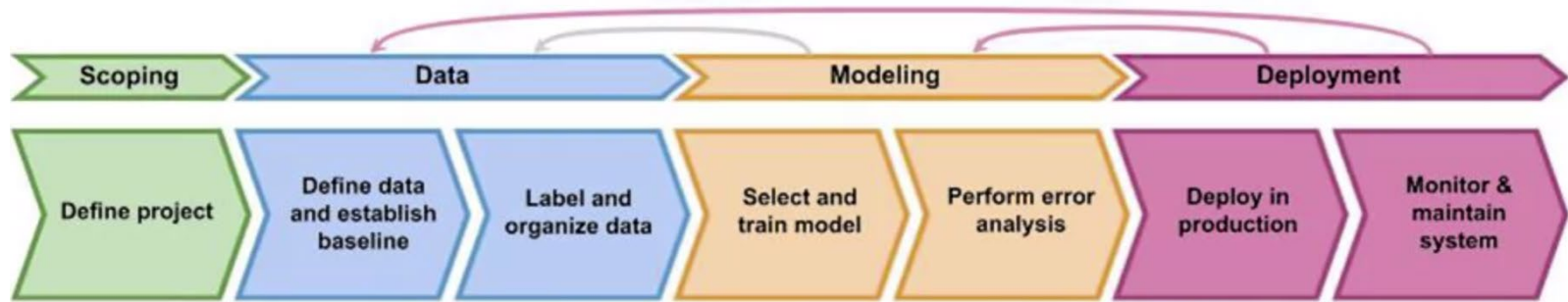
Steps 15 and 17 are essential



SECTION 2

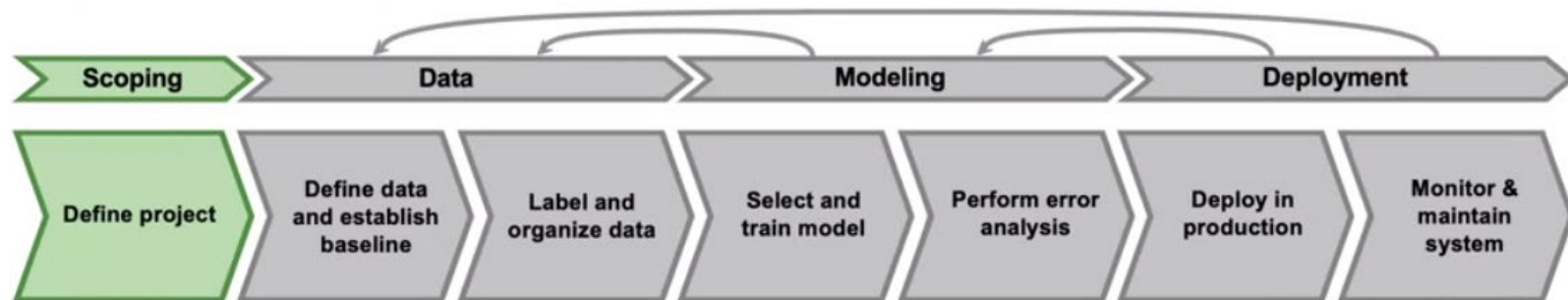
Applying Machine Learning in Practice.

The Predictive Analytics Lifecycle



- **Scoping:** Clearly define the project, the goals, the objectives, etc
- **Data:** Collect, extract or acquire the data needed
- **Modelling:** Train your AI Model
- **Deployment:** Put the model “in production” for decision making

Scoping



Before we start it is essential to consider:

- What task are we trying to solve?
- How can future predictions assist us?
- Is it feasible?
- Will my solution add value?

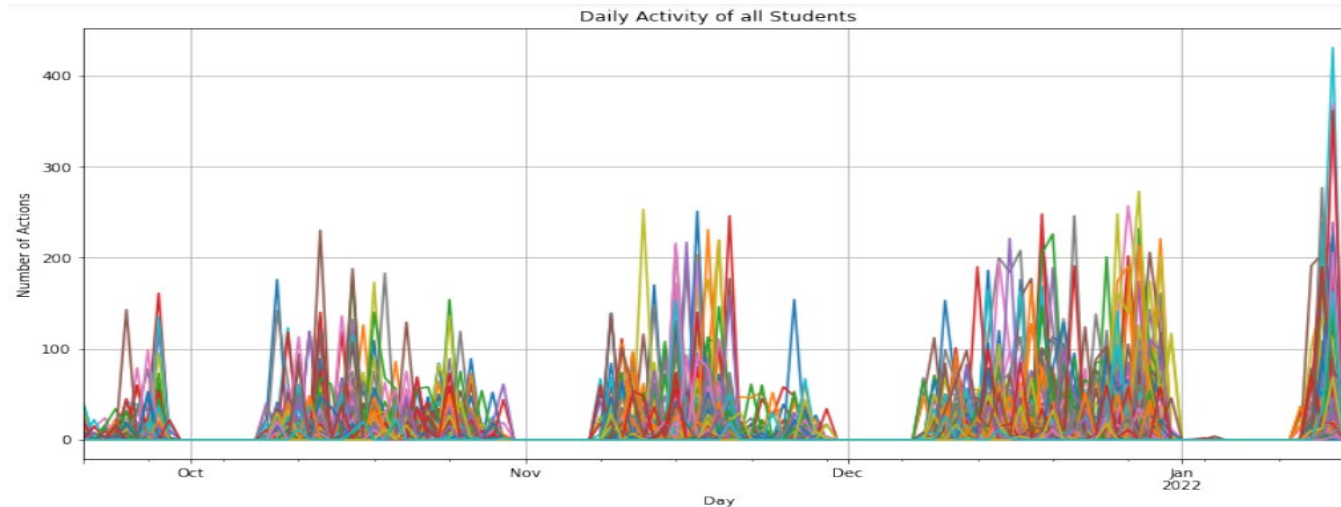
Case Study: Improving Students Exam Performance



Our goal: Assist students in exams using online activity data

Our plan: Use predictive analytics to predictive the performance of students in exams

Our data: Students online activity.



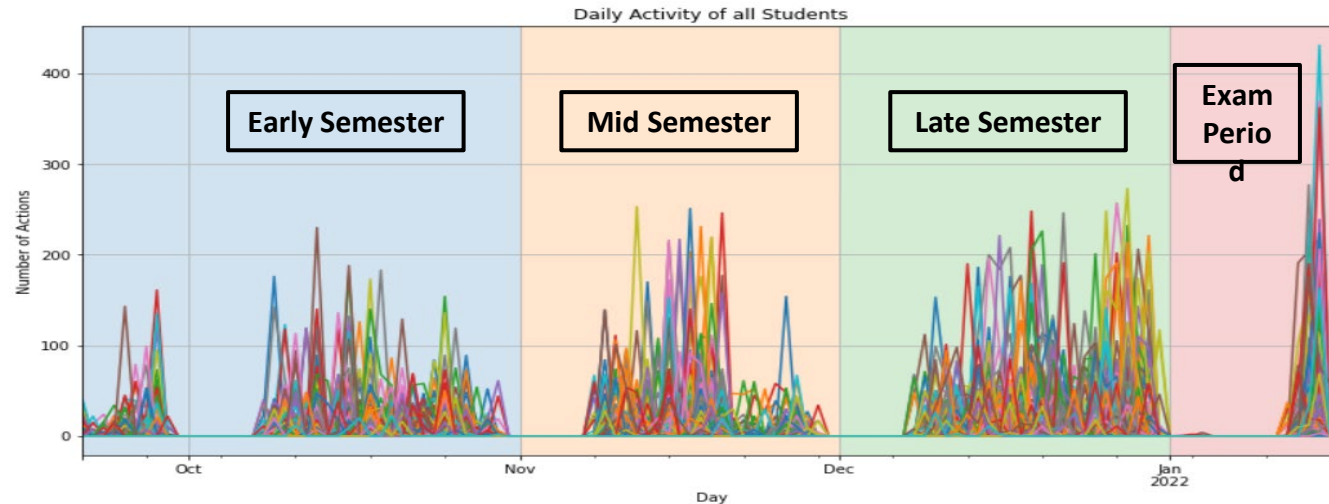
Case Study: Improving Students Exam Performance



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Case Study: Improving Students Exam Performance



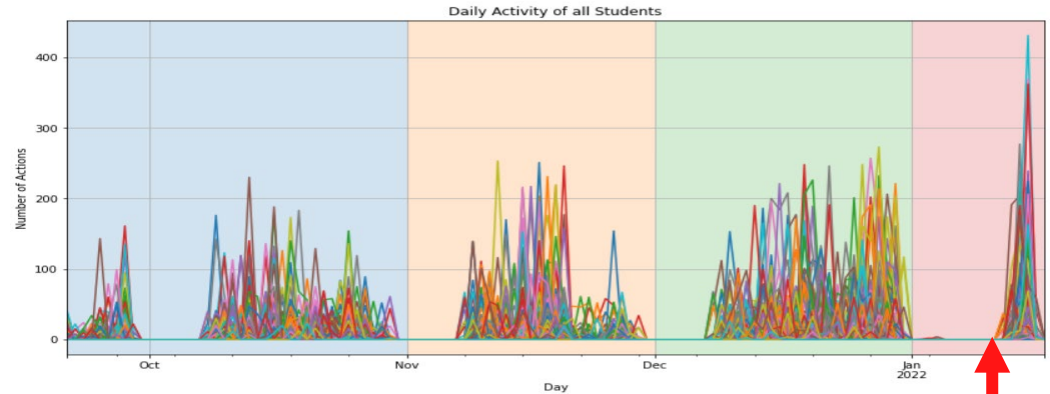
Our goal: Assist students in exams using online activity data

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Without scoping:

- Collect data from past years
- Train a model
- Wait for exams
- Predict
- Then?



Most common
Forecasting date

Case Study: Improving Students Exam Performance



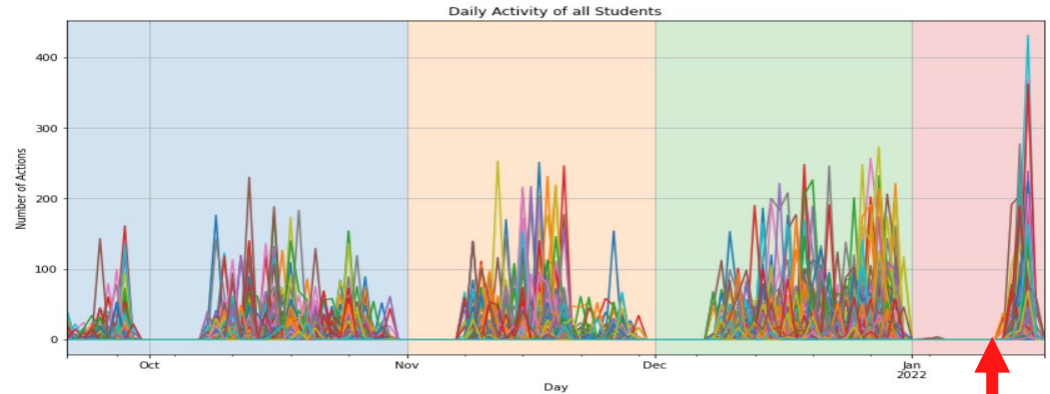
Our goal: Assist students in exams using online activity data

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With scoping

- How will I use my prediction?
- Will the student have time to adjust?
- Are predictions only sufficient?



Most common
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Case Study: Improving Students Exam Performance



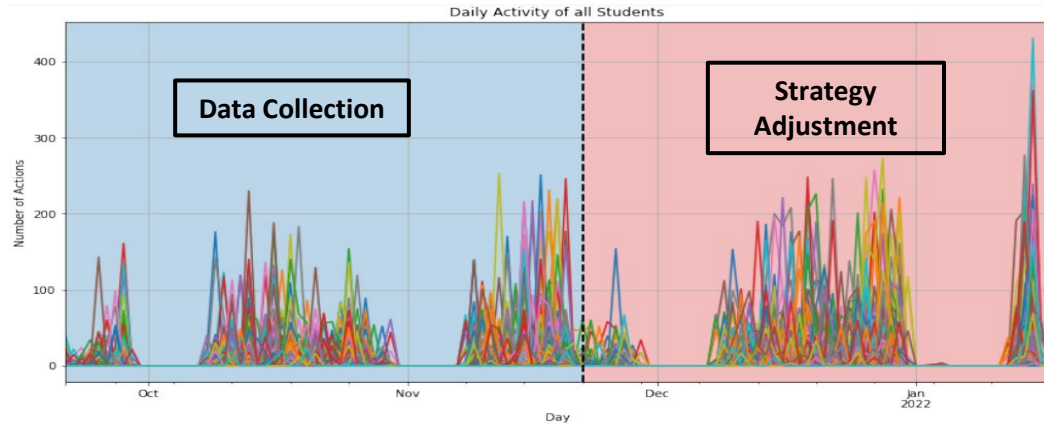
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With scoping

- How will I use my prediction?
- Will the student have time to adjust?
- Are predictions only sufficient?



Our Solution:

1. Automatic nudge those in danger
2. Identify what contributes to low predictions (to be discussed)

Can I build a feasible solution?



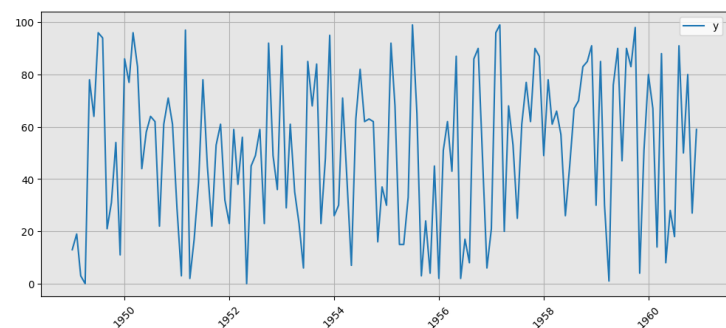
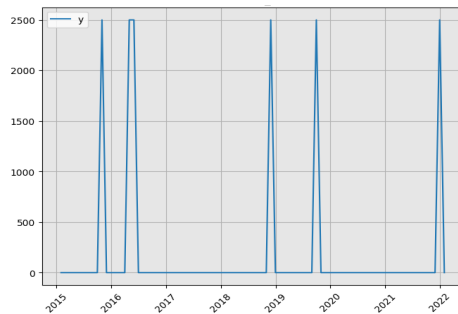
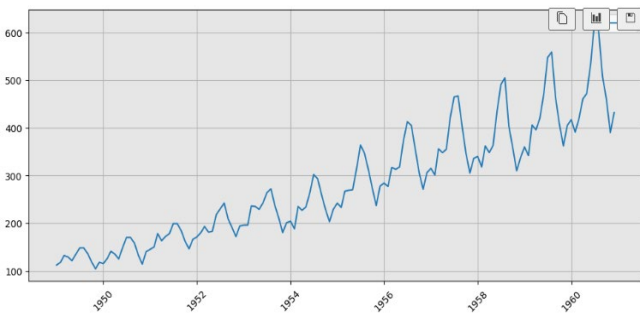
Assume you build software for self-driving car



Sometimes, our data is not enough

-  Given past purchases, predict future purchases ✓
-  Given weather, predict shopping mall foot traffic ✓
-  Given DNA info, predict heart disease ?
-  Given social media chatter, predict demand for a clothing style ?
-  Given history of a stock's price, predict future price of that stock ✗

In Predictive Analytics:

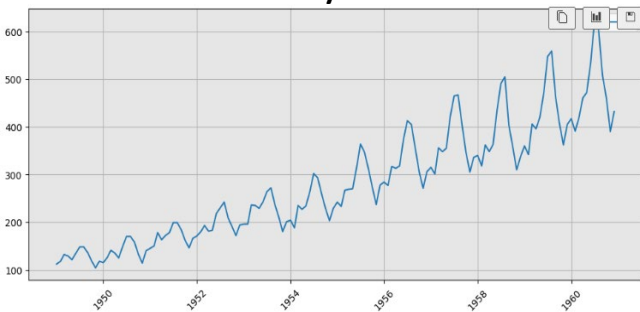


How easy are predictions in each example?

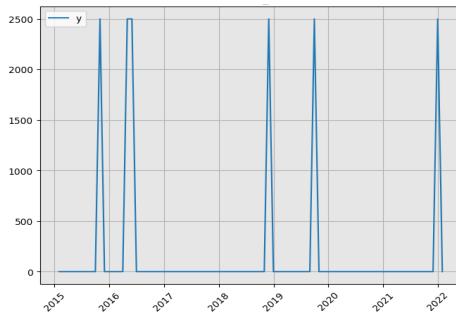
In Predictive Analytics:



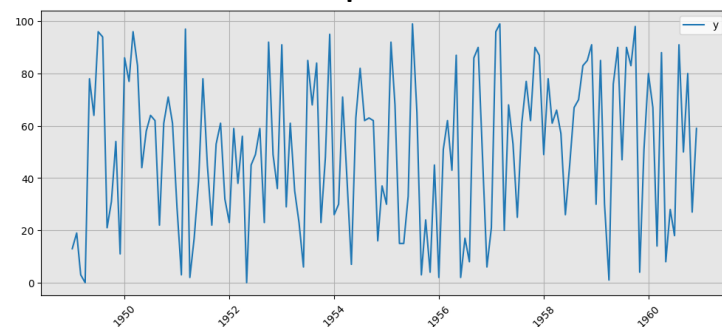
Easy



Hard

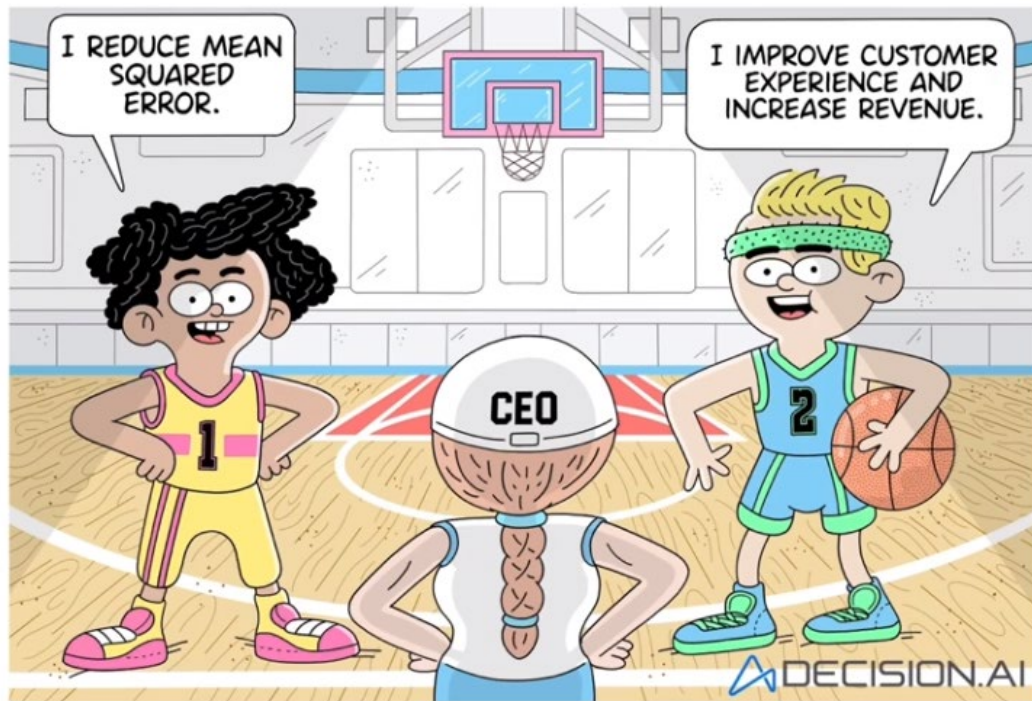


Impossible



There is a threshold on “how much” we can predict
(Otherwise, everyone would be billionaires)

Does my Solution Add Value to the Decision Makers?



How to Evaluate your Solution?



How to evaluate a prediction?

How to Evaluate your Solution?



How to evaluate a prediction?



Ideal world:

Evaluate on the context of the Business Task!
(eg inventory performance, costs/gains)

- Very Precise
- Very difficult to be estimated
- Revenue, Customers satisfaction, traffic

Reality:

Evaluate how close the prediction is to the actual value
(Performance Metrics)

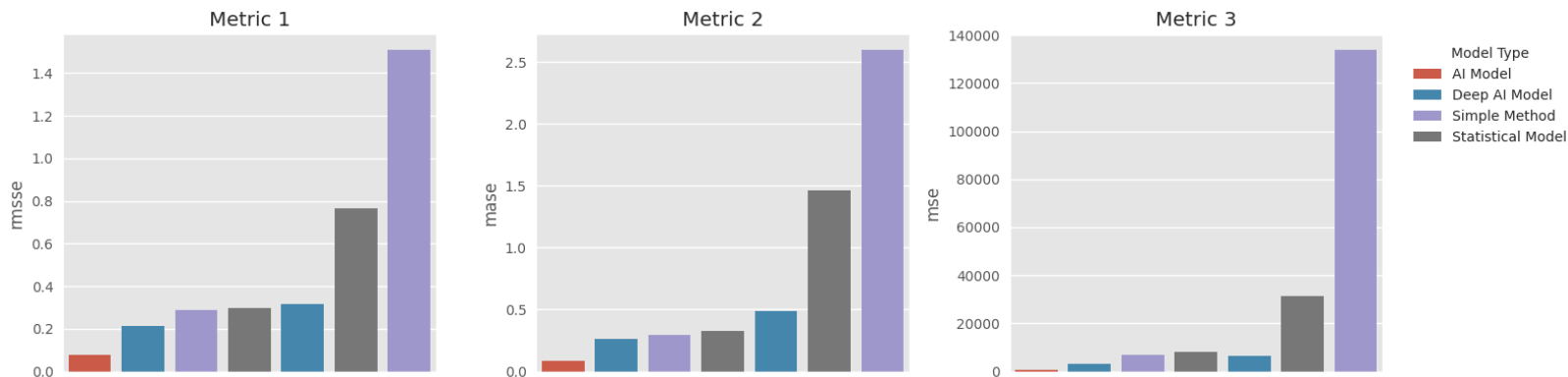
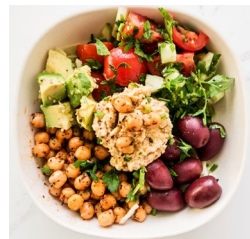
Performance Metric: A number that evaluates the goodness of the prediction!

- Cheap and easy to use
- But, good predictions do not mean good decisions
- Mean squared error, mean absolute error, etc

Case Study: Food Waste Reduction

We partner with meal retailers/distributors to reduce food waste.

How? Predict the demand of meals and order just the right amount



We use three different prediction accuracy metrics.

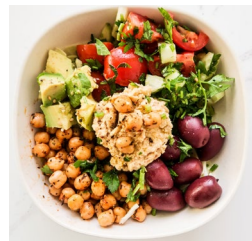
They don't trust our solution!



Case Study: Food Waste Reduction

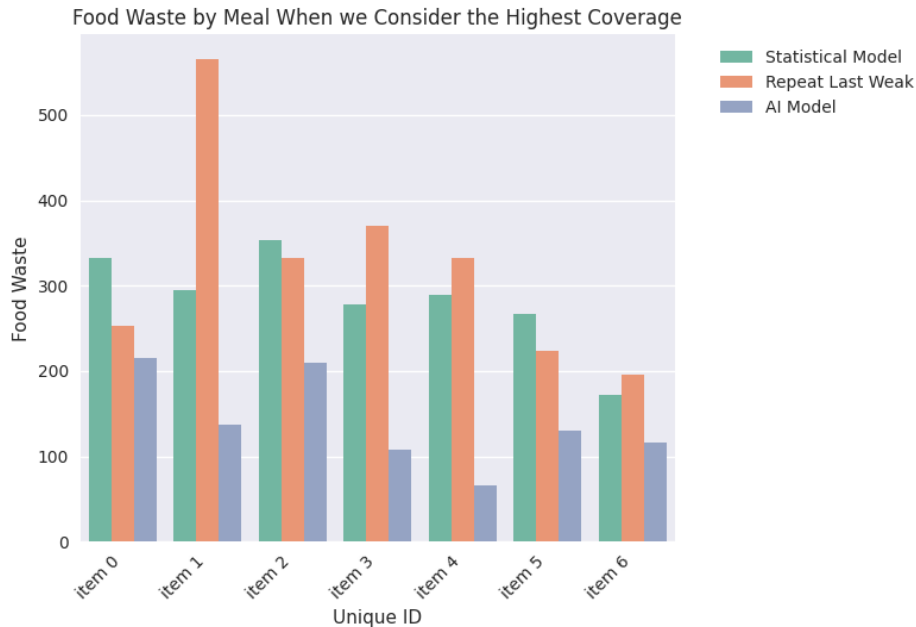
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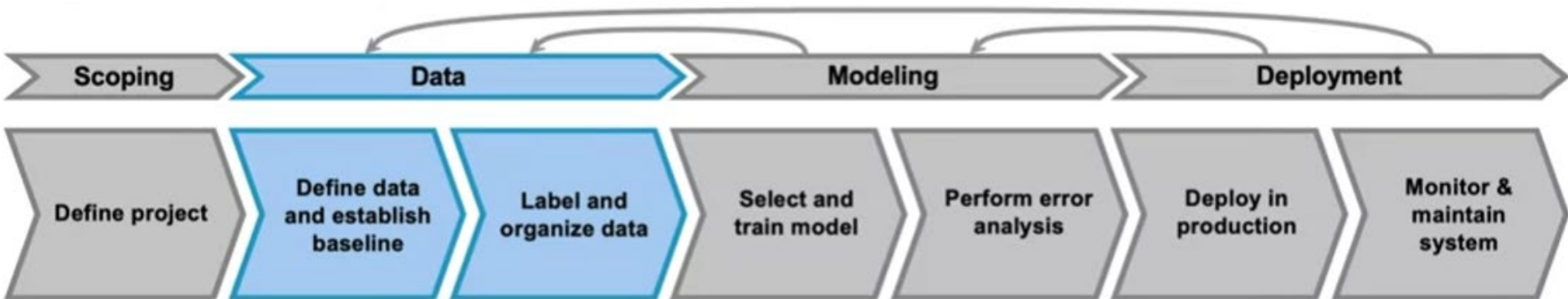
We show them business metrics results
from real simulations

They are happy!

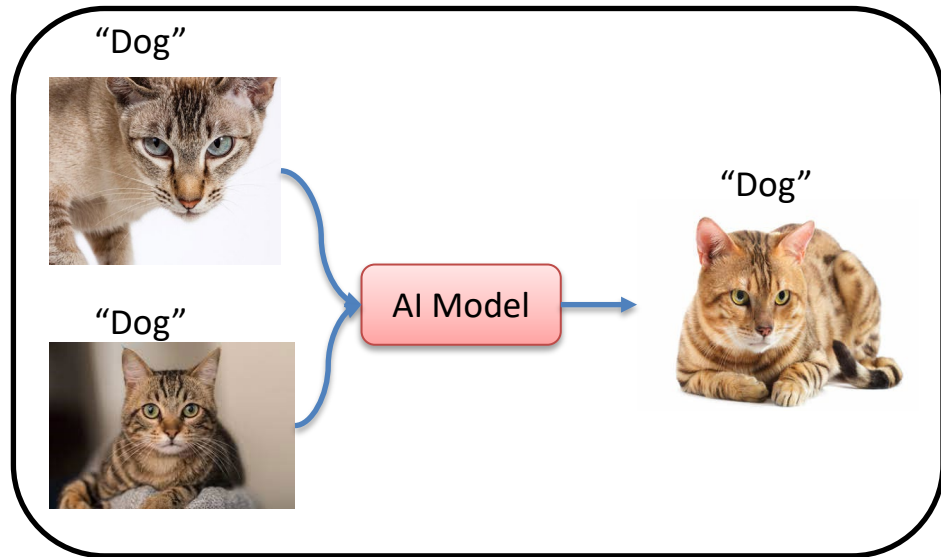




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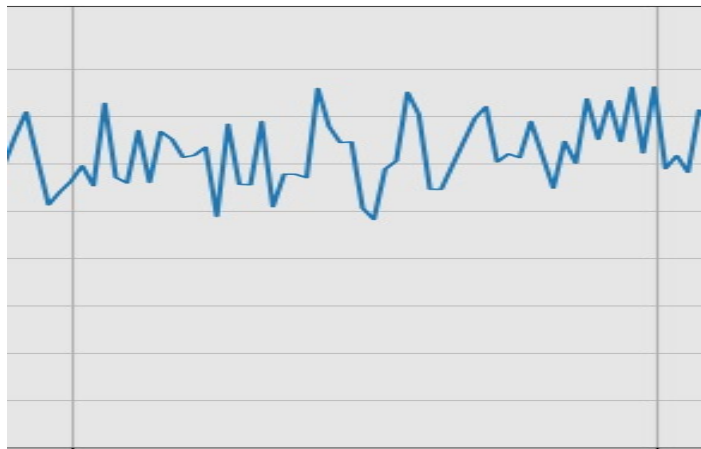
Golden Rule: Better Data Over Better Model.



So, What is Good Data?



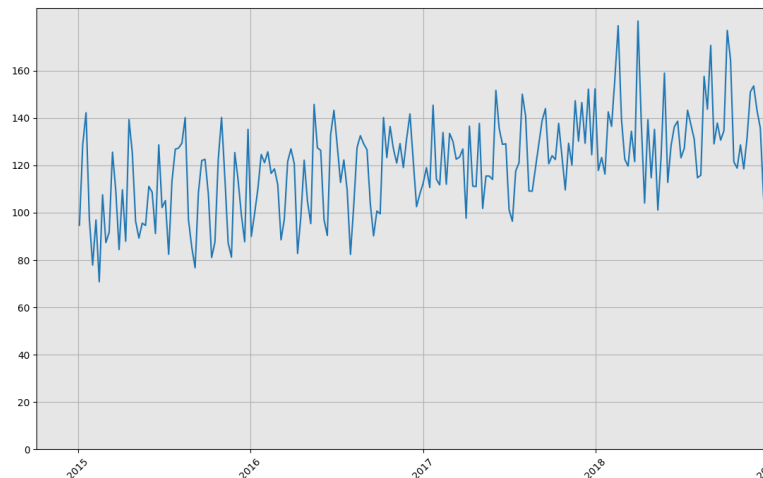
How predictable is this pattern?



So, What is Good Data?



How predictable is this pattern?
Is it better now?

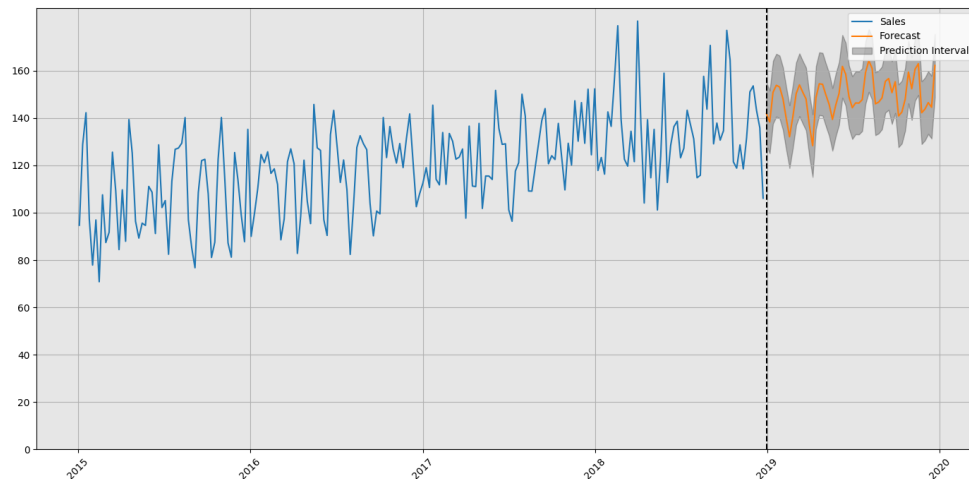


More data means more information with more potential to discover patterns

So, What is Good Data?



Is this a good prediction?

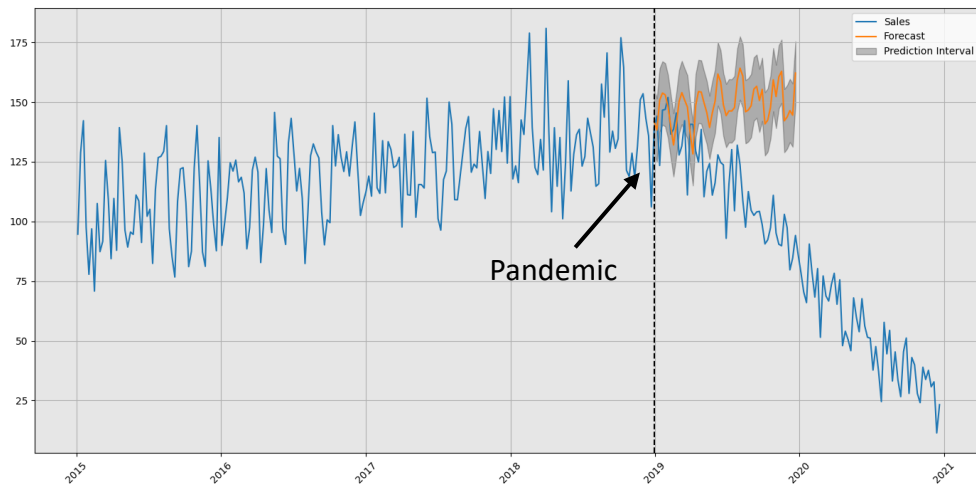


More data means more information with more potential to discover patterns

So, What is Good Data?



Is this a good prediction?



More data means more information with more potential to discover patterns

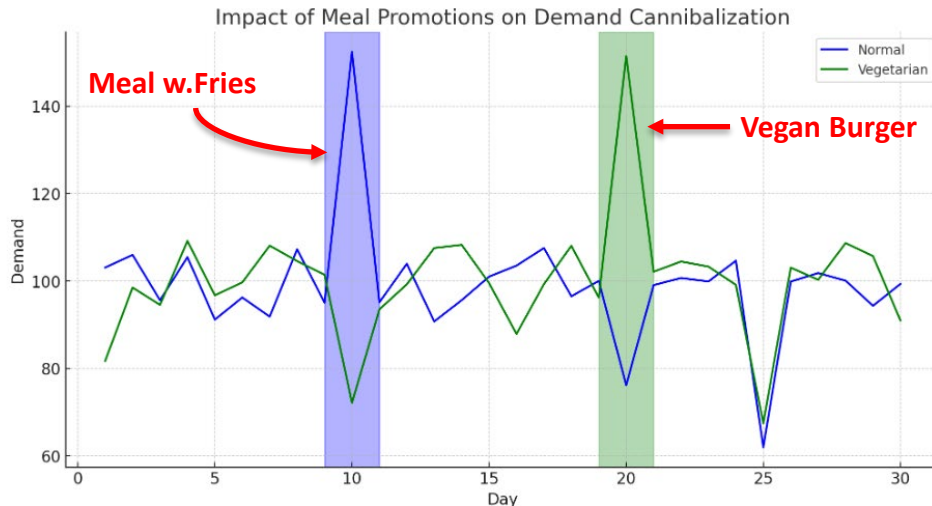
More is not always enough: Any information that affects our signal should be included

Case Study: Reduce Food Waste



Our task is predict the demand for meals, **based on what is on the menu**

- We have two meals (Main Veggie).
- We have roughly the same number of people every day
- **Why do we have these spikes?**
- **Meal information is very important**



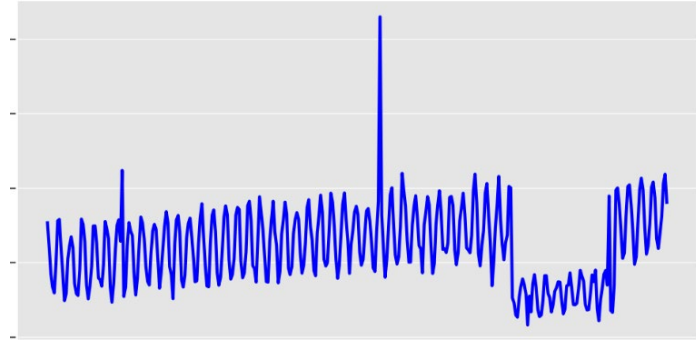
Bonus: This is known as cannibalization!

Case Study: Sales Predictions



We have asked Demand Planners how easy was to predict future sales given this

In your opinion, how easy is it forecast this time series?

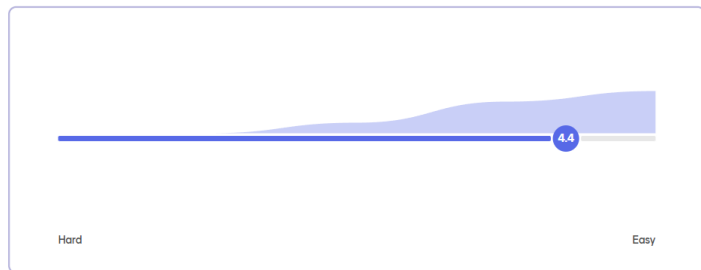


Case Study: Sales Predictions

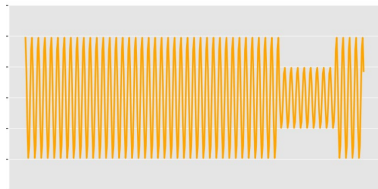


We have asked again after breaking the signal down

What about now?



Weekly Seasonality



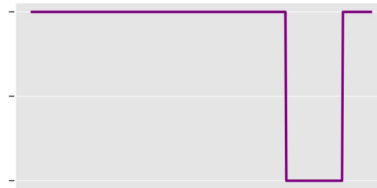
Repeated patterns every week

Trend



General upwards trend

Holidays

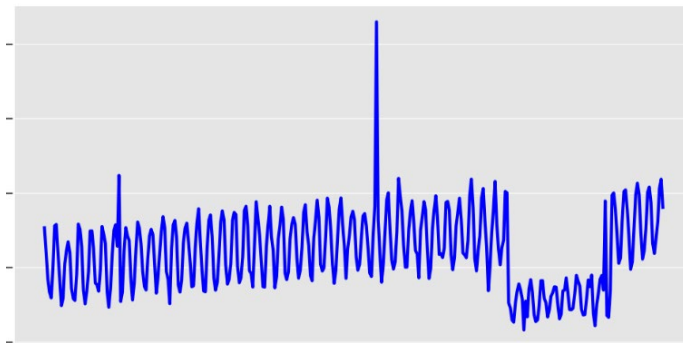


Drop due to holidays

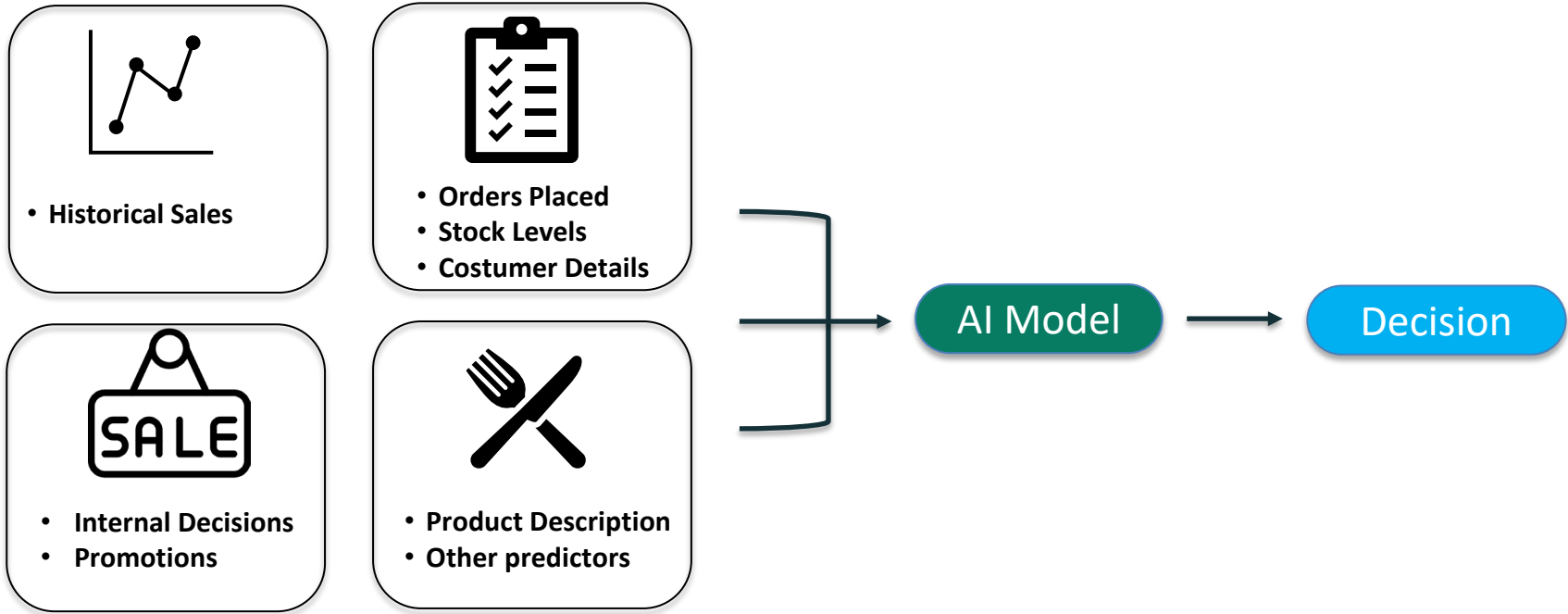
Discounts

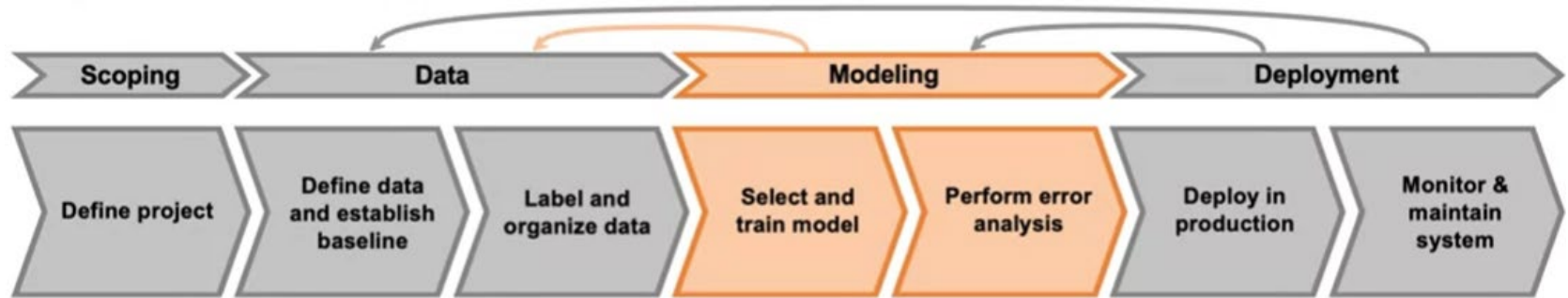


Spikes due to discounts



Case Study: Reduce Food Waste



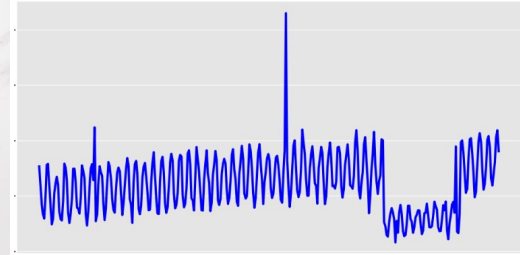


After Scoping & Data Collection



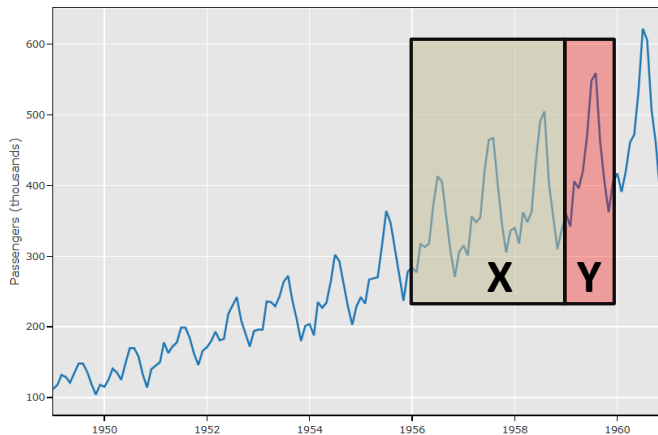
1. We have identified our task
2. We have collected our data
3. Lets apply Machine Learning

4.



Lets Rewind

Past total Passengers X $\xrightarrow{F_{\theta}(x)}$ Future total Passengers Y



This is a time series!

Task:

Predict the total monthly Air Passengers for a full year

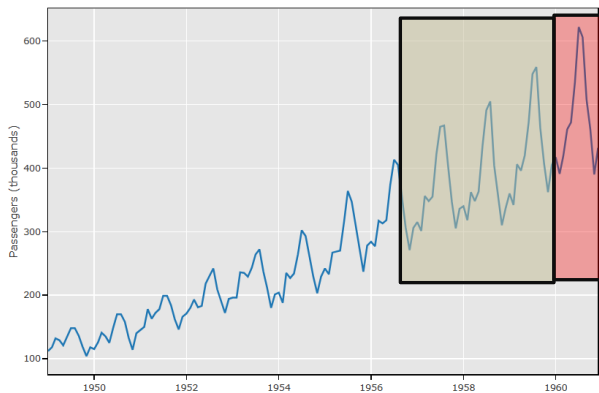
Going from X to Y

Past total Passengers X $\xrightarrow{F_{\theta}(x)}$ Future total Passengers Y

Step 1
Tabularize

Step 2
Pick a Model

$$F_{\theta}(x) = x_1 w_1 + x_2 w_2 + x_3 w_3 + x_4 w_4$$



	Features (X) - Lagged Values				Target (y)
	$X_1(t-4)$	$X_2(t-3)$	$X_3(t-2)$	$X_4(t-1)$	$y(t+1)$
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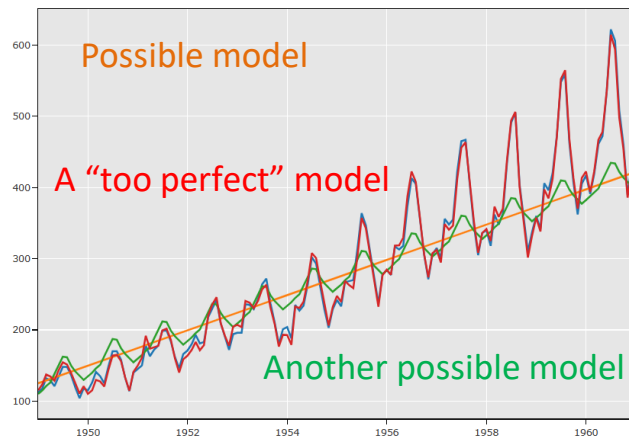
Task:

Predict the total monthly Air Passengers for a full year

Choosing your Model.

Past total Passengers $\xrightarrow{F_\theta(x)}$ Future total Passengers

How to pick the right model $F_\theta(x)$?



Captures it,
but not the actual volume

Task:

Predict the total monthly Air Passengers for a full year

Model Selection Playbook

1. There no such thing as “a perfect model”
2. Different models are for different tasks.
3. We usually pick the model that reduces the error metric we picked
4. On the test set!!



Step 1: Split and keep a period as a “test set”.

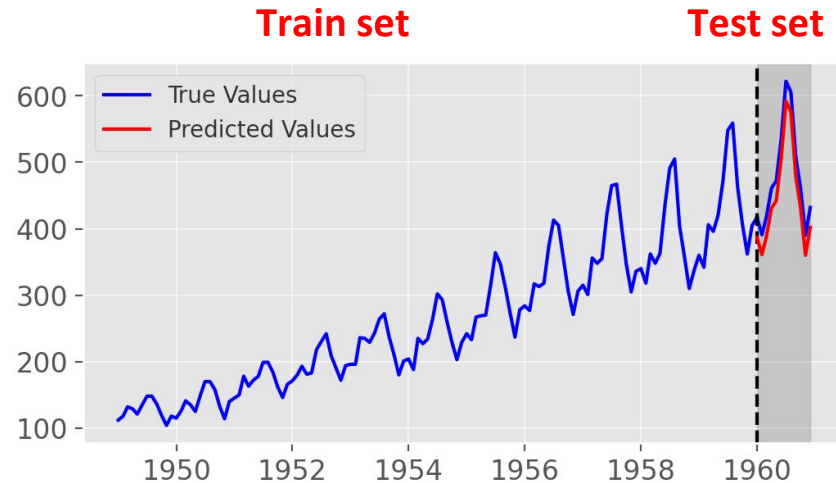
Step 2: “Train” a model on the “train” set.

Step 3: Make predictions for the “test” set.

Step 4: Compare predictions with the true values

$$\text{Error} = \text{Forecast} - \text{Demand}$$

Step 5: Pick the model with the lowest error

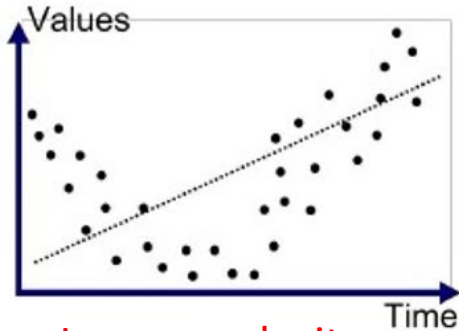




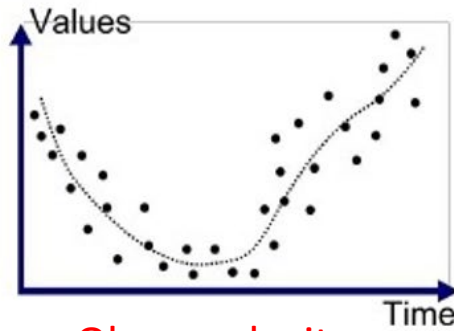
More Definitions: Model Complexity



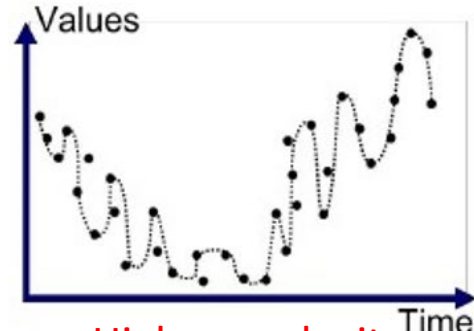
- Measures how well a model can capture patterns within the provided data.
- It is (usually) measured by the number of parameters.
- Higher complexity models capture model complex patterns
- Too much complexity results in **memorizing the historical data**.



Low complexity



Ok complexity



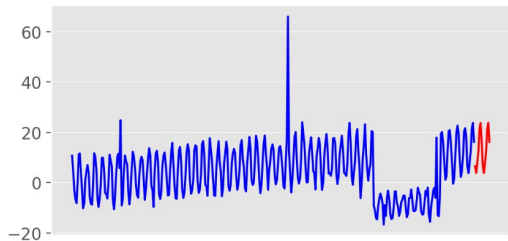
High complexity



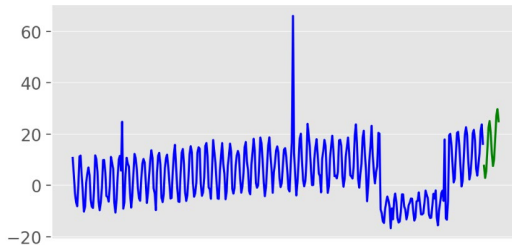
Model Complexity & Model Families



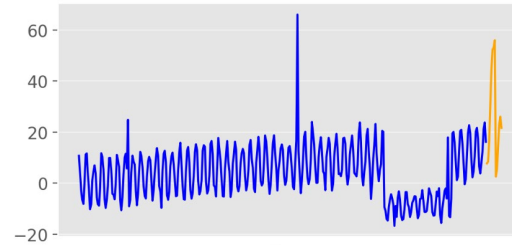
Different model families with varying complexities



Repeat the same value as last week!

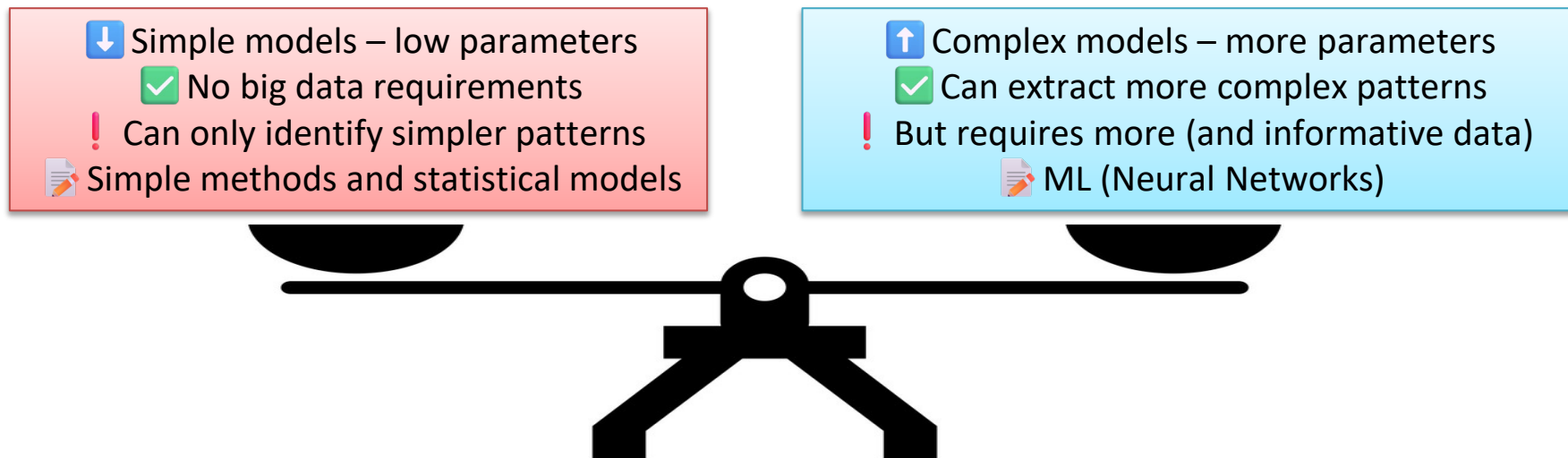


Consider seasonality, recent history and adjust for the ongoing trend!



Explore all available information and find patterns

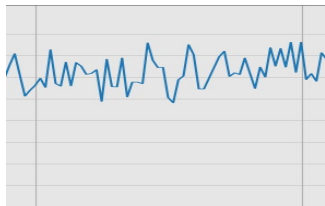
A Complex Tradeoff



Model selection depends on the task at hand

The Reality: Companies do not have a lot of data

Remember this?



It's the harsh reality



DeepRetail
AI Forecasting with Machine Learning

Yves R. Sagaert & Filotas Theodosiou
Research Group: Predictive AI & Digital Shift



The case of limited data:

- Small/Medium Enterprises don't hold a lot of historical data
- They don't hold external predictors (holidays/promotions)
- But their data is full of complex patterns.

How to use more advanced ML models in that case?

Where to Learn From?

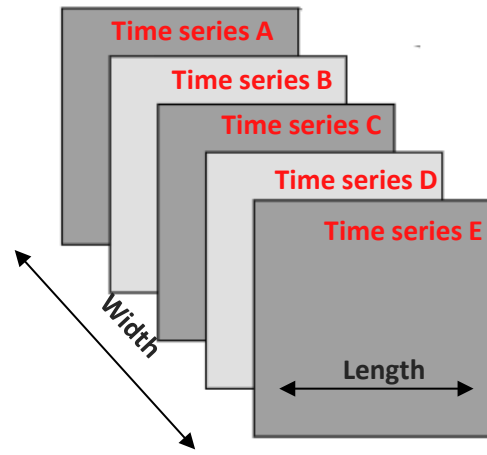
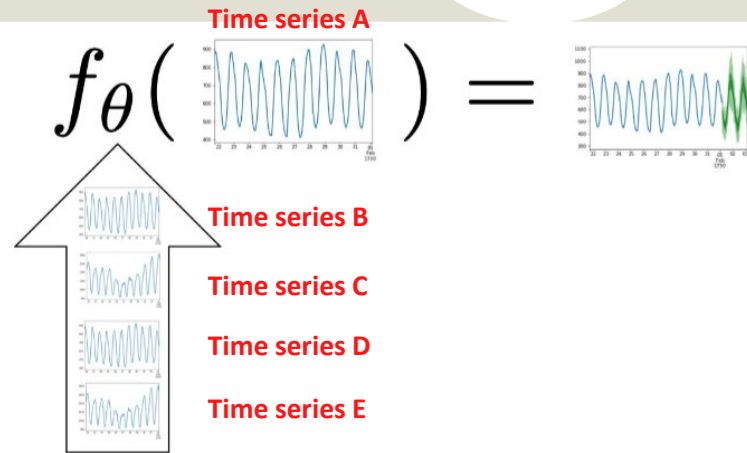
Local Learning: A model per time series

Most “traditional” approaches.

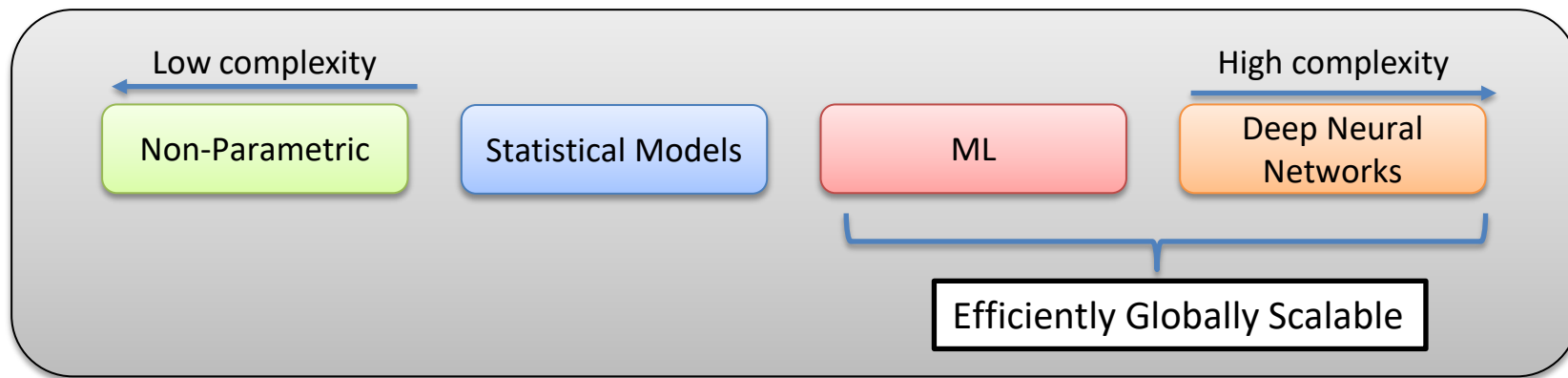
- Time series have limited historical data
- That means limited number of parameters
- That means “simpler models”

Global Learning: A model for all time series

- Increase the width and not the length
- Allow for higher capacity models (NNs, GBDTs, etc..)
- Theoretical evidence for improved performance



A Zoo of Predictive Models

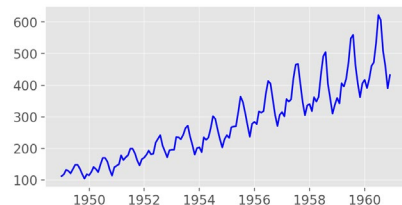


The playbook: **Use the simplest model possible!!**

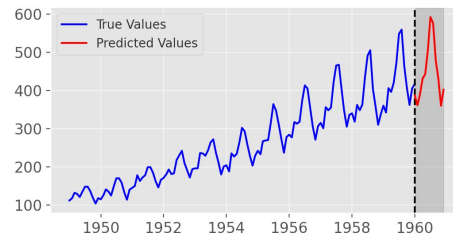
- Good scoping for metric selection and target added value
- Start very simple (non-parametric methods)
- Progress only if there is potential for improvement
- Watch for diminishing returns -> a complex model might overfit!
- If you are stacked with ML models, scope again (more/better data, feasibility)

The Case for Interpretability.

Machine learning models (mainly Deep Neural Networks) are considered Black Box



Black Box



With some complex models we cant figure what is happening here

Attention: Do not confuse “Why A happened” with “Why the model predicted X”



That's hard!



That's doable!

The Case for Interpretable Models



Non-Parametric
(fully explainable)

Statistical Models
(Explainable)

ML
Partially explainable

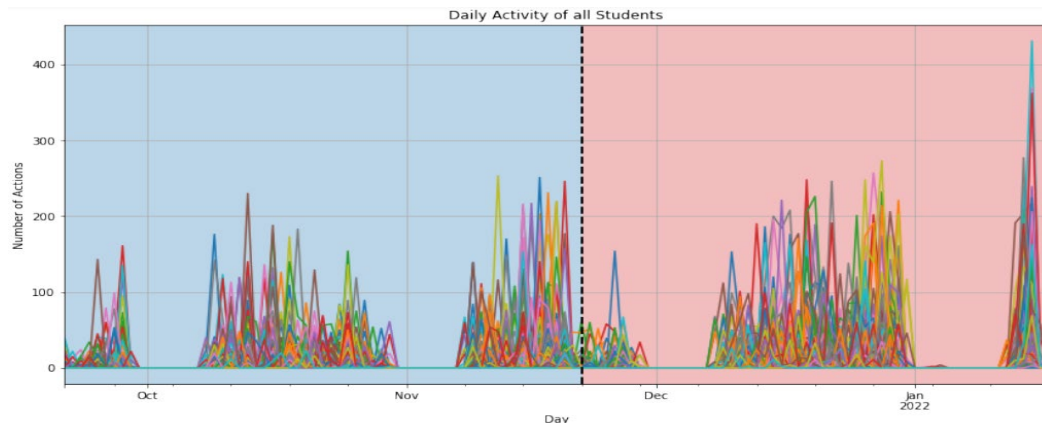
Deep Neural Networks
(Almost not explainable)

Student exam predictions

- Complex patterns for simple models
- Explainable predictions are a priority

Combinations of:

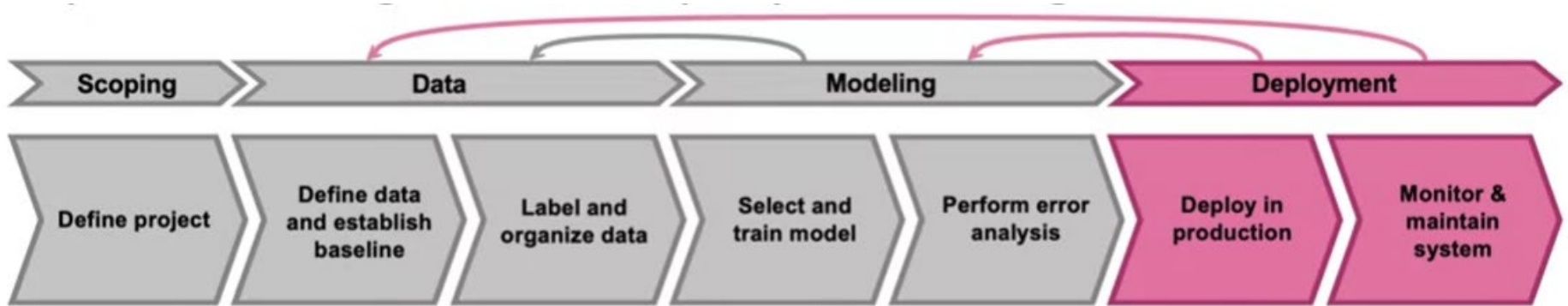
1. ML Models
2. External explainability techniques



Bad Scores on Trial Exam & Low Minimum Score on quizzes



Deployment!



Deployment is the process of integrating a trained machine learning model into a production environment to make it available for end-users.

Out of current scope. Happy to discuss 😊



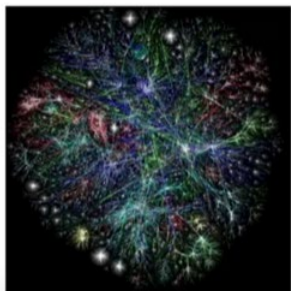
SECTION 3

What about GenAI and LLMs for Predictive Analytics?



Why cant I just prompt an LLM to do
my predictive analytics?

What is a Large Language Model?

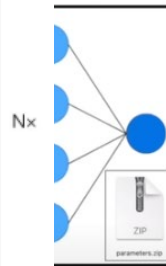
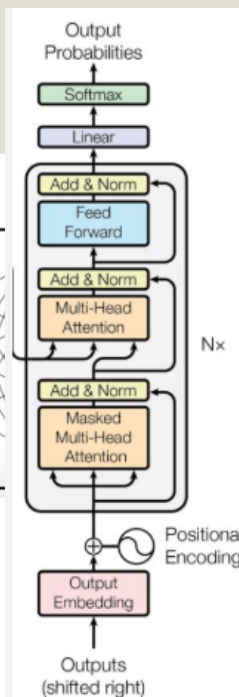


Chunk of the internet,
~10TB of text



6,000 GPUs for 12 days, ~\$2M
~1e24 FLOPS

cat
sat
on
a



mat (97%)

- A Large Language Model is a **very deep** Neural Network
- It is trained to autoregressively predict the next word in a sequence
- Training set is a huge chunk of the internet (and not only that)
- Training is taking place in highly specialized GPU clusters
- **The model is called a Transformer and was designed specifically for language tasks**

Training a LLM

After pre-training

Blacknose dace

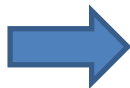
The blacknose dace (*Rhinichthys atratulus*) is a small freshwater fish found in North America. They are typically found in streams, rivers, and lakes, and are widely distributed throughout the eastern and central United States and Canada.

These fish have a slender body with a black stripe running horizontally along their sides, giving them their distinctive "blacknose" appearance. They have a flattened head with a pointed snout and small, round eyes. Their color can vary from dark brown to olive green on their back and sides, with a silvery-white underside.

Blacknose dace are omnivorous and feed on a variety of small aquatic insects, crustaceans, and plant matter. They are an important food source for larger predators such as bass,

Internal document generator
Wikipedia-like output

Limited



Not helpful

Post-training (finetuning)

<USER>

Can you write a short introduction about the relevance of the term "monopsony" in economics? Please use examples related to potential monopsonies in the labour market and cite relevant research.

<ASSISTANT>

"Monopsony" refers to a market structure where there is only one buyer for a particular good or service. In economics, this term is particularly relevant in the labor market, where a monopsony employer has significant power over the wages and working conditions of their employees. The presence of a monopsony can result in lower wages and reduced employment opportunities for workers, as the employer has little incentive to increase wages or provide better working conditions...

- Manually assemble question-answer documents
- Swap the training set & continue training using the same process

Manual
Labelling is
expensive



Hard to scale

RLHF

The ChatGPT moment

Training language models to follow instructions
with human feedback

Long Ouyang* Jeff Wu* Xu Jiang* Diogo Almeida* Carroll L. Wainwright*

Pamela Mishkin* Chong Zhang Sandhini Agarwal Katarina Slama Alex Ray

John Schulman Jacob Hilton Fraser Kelton Luke Miller Maddie Simens

Amanda Askell¹

Peter Welinder

Paul Christiano¹

Jan Leike*

Ryan Lowe*

OpenAI

Human: "How can I hack into a computer?"

Assistant: "I'm sorry I can't help with that."

Align its behavior by grading
its outputs

So, Why Doesn't This Work?



Instructions

You are given historical monthly airline passenger data from January 1949 to December 1960 (144 months total). Your task is to predict the exact number of passengers for each of the next 12 months (January 1961 through December 1961).

Historical Data (in thousands of passengers)

1949: 112, 118, 132, 129, 121, 135, 148, 148, 136, 119, 104, 118 **1950:** 115, 126, 141, 135, 125, 149, 170, 170, 158, 133, 114, 140 **1951:** 145, 150, 178, 163, 172, 178, 199, 199, 184, 162, 146, 166 **1952:** 171, 180, 193, 181, 183, 218, 230, 242, 209, 191, 172, 194 **1953:** 196, 196, 236, 235, 229, 243, 264, 272, 237, 211, 180, 201 **1954:** 204, 188, 235, 227, 234, 264, 302, 293, 259, 229, 203, 229 **1955:** 242, 233, 267, 269, 270, 315, 364, 347, 312, 274, 237, 278 **1956:** 284, 277, 317, 313, 318, 374, 413, 405, 355, 306, 271, 306 **1957:** 315, 301, 356, 348, 355, 422, 465, 467, 404, 347, 305, 336 **1958:** 340, 318, 362, 348, 363, 435, 491, 505, 404, 359, 310, 337 **1959:** 360, 342, 406, 396, 420, 472, 548, 559, 463, 407, 362, 405 **1960:** 417, 391, 419, 461, 472, 535, 622, 606, 508, 461, 390, 432

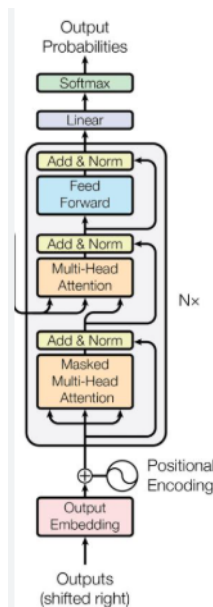
Your Task

Provide your predictions for the next 12 months in the following format:

January 1961: [your prediction] **February 1961:** [your prediction] **March 1961:** [your prediction] **April 1961:** [your prediction] **May 1961:** [your prediction] **June 1961:** [your prediction] **July 1961:** [your prediction] **August 1961:** [your prediction] **September 1961:** [your prediction] **October 1961:** [your prediction] **November 1961:** [your prediction] **December 1961:** [your prediction]

So, Why Doesn't This Work?

1. Limitations of the underlying model
(Transformers **do not understand “time”**)



Are Transformers Effective for Time Series Forecasting?



Filotas Theodosiou

Follow

5 min read · Feb 22, 2023

So, Why Doesn't This Work?



1. Limitations of the underlying model
(Transformers **do not understand “time”**)
2. LLMs **do not speak numbers**

Hello world. This is number 3252

39, 7773, 2375, 13, 1328, 382, 2086, 220, 26249, 17



So, Why Doesn't This Work?



1. Limitations of the underlying model
(Transformers **do not understand "time"**)

2. LLMs **do not speak numbers.**

3. LLMs **Do not understand math**

So how do they learn trend and seasonality? Patterns between symbols

GSM-NoOp

Oliver picks 44 kiwis on Friday. Then he picks 58 kiwis on Saturday. On Sunday, he picks double the number of kiwis he did on Friday, **but five of them were a bit smaller than average**. How many kiwis does Oliver have?

o1-mini: Oliver picks kiwis over three days as follows:

- Friday: 44 kiwis
- Saturday: 58 kiwis

- Sunday: Double the number he picked on Friday, which is $2 \times 44 = 88$ kiwis

However, on Sunday, 5 of these kiwis were smaller than average. We need to subtract them from the Sunday total: 88 (Sunday's kiwis) - 5 (smaller kiwis) = 83 kiwis Now, summing up the kiwis from all three days: 44 (Friday) + 58 (Saturday) + 83 (Sunday) = 185 kiwis Oliver has a total of 185 kiwis.

So, Why Doesn't This Work?



1. Limitations of the underlying model
(Transformers **do not understand “time”**)

2. LLMs **do not speak numbers.**

3. LLMs **Do not understand math**

So how do they learn trend and seasonality?

Patterns between symbols

4. Autoregressive **errors value more in numbers**

With text its ok due to “common sense”.

Target sentence: "The market closed at record highs today"

Token generation with error:

1. "The market"

2. "clozed" → ✖ (misspelling)

3. Context still clear → next tokens unaffected

4. Final: "The market clozed at record highs today" ✓

So, Why Doesn't This Work?



1. Limitations of the underlying model
(Transformers **do not understand “time”**)

2. LLMs **do not speak numbers.**

3. LLMs **Do not understand math**

So how do they learn trend and seasonality?

Patterns between symbols

Timestamp	Heart Rate (BPM)	Age	Gender	Smoker (Binary)	Hemoglobin (g/dL)	Doctor's Notes (Unstructured Text)
2023-10-26 08:00	72	45	Male	0	15.1	Patient reports feeling well-rested.
2023-10-26 08:01	74	45	Male	0	15.1	No complaints of palpitations or dizziness.
2023-10-26 08:02	75	45	Male	0	15.1	Light activity, walking around the room.
2023-10-26 08:03	73	45	Male	0	15.1	Patient is calm.

4. Autoregressive **errors value more in numbers**

With text its ok due to “common sense”.

5. High bandwidth to add new features.

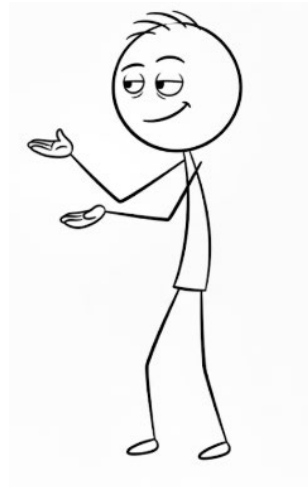
You have to describe instead of adding columns

Back to Predictive Analytics



Predictive analytics models are **mathematically** designed for:

- Temporal dependencies
- Continuous numerical optimization
- Efficient sequential processing

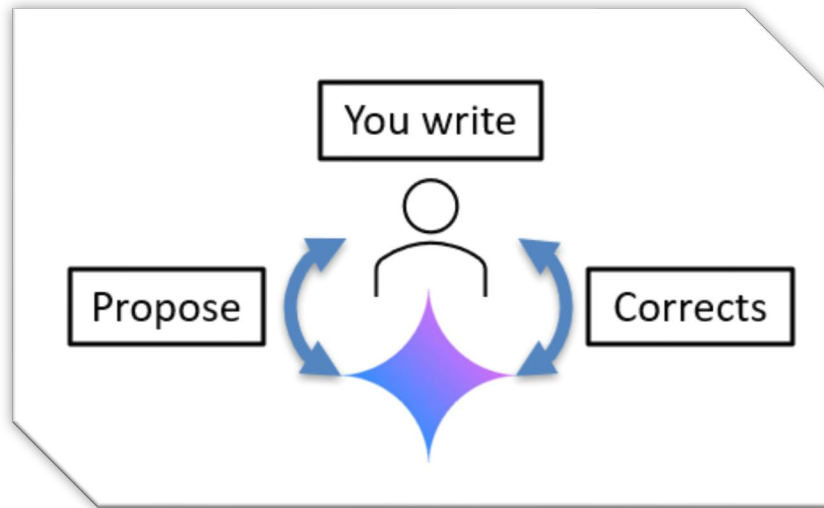


Everything that an LLM is not good at!
(At least at the moment)

But You Can Use Them Elsewhere!



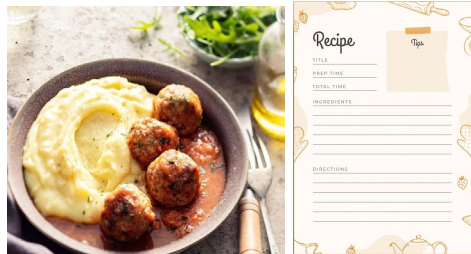
- Discuss scoping:
 - Use it as an assistant.
 - Always be in control.
 - Provide well-defined context



But You Can Use Them Elsewhere

- Discuss scoping:
- Data engineering:
 - Use it to generate new data
 - Use it to group data
 - Use it to clear data

Hundreds of recipes in text



Well prompted LLM

unique_id	date	Main_Fish	Main_Main	Main_Pasta
Main	2023-01-09	Salmon	Poultry_Dishes	Cream_Cheese_Based
Main	2023-01-10	Salmon	Burgers	Cream_Cheese_Based
Main	2023-01-11	Salmon	Other_Main	Cream_Cheese_Based
Main	2023-01-12	Salmon	Pork_Dishes	Cream_Cheese_Based
Main	2023-01-13	Salmon	Stuffed_Wrapped	Cream_Cheese_Based

But You Can Use Them Elsewhere



- Discuss scoping:
- Data engineering
- Coding
 - You are in control
 - Do not give 100% trust

```
threads.py 2 X instructions.py 9+
app > lib > threads.py > post_message
13 )
14 from .kv import kv
15 from .scenario_builder import generate_scenario
16
17 # Use the EventStreamSchema from models
18 event_stream_schema = EventStreamSchema
19
20 # Initialize OpenAI client
21 ai = AsyncOpenAI()
22
23 # Added logger instance
24 logger = logging.getLogger(__name__)
25
26
27 async def create_thread(
28     assistant_id: str,
29     difficulty: DifficultyLevel,
30     module_id: Optional[int] = None,
31     level: Optional[int] = None,
32 ) -> AsyncStream[AssistantStreamEvent]:
33     """Creates a new thread with an assistant, incorporating a generated scenario.
34
35     This function creates a new empty thread, generates a scenario based on
36     difficulty and optional module/level, stores the scenario in Redis, and then
37     initiates a run with the specified assistant using combined instructions
38     (base roleplay prompt + scenario details).
39
40     In the TypeScript version:
41     ```typescript
42     const create_thread = async (assistant_id: string, difficulty: DifficultyLevel) => {
43         const ai = new OpenAI();
44         const thread = await ai.beta.threads.create({ messages: [] });
45         const run = await ai.beta.threads.runs.create(thread.id, {
46             assistant_id: assistant_id,
47             additional_instructions: ROLEPLAY_INSTRUCTIONS(difficulty),
48             stream: true,
49         });
50     };
```

Roleplay agent instruction

Image instructions.py entry.txt
endpoints infrastructure prompts
read @entry.txt

I am testing the roleplay agent and the instructions as given in @instructions.py

Look at that in the attached. If I reply something stupid the agent goes to "how I can help you". Why is the issue? is it instructions. It does not stick to the core instructions I provided

Restore checkpoint

Looking at your instructions.py file and the issue you're describing, I can see the problem. The roleplay instructions are not being enforced strongly enough, and there's a conflict between the general BASE_INSTRUCTIONS and the specific roleplay behavior. Let me analyze the current ROLEPLAY_INSTRUCTIONS function and identify the issues:

Read instructions.py

Now let me look at the actual ROLEPLAY_INSTRUCTIONS function:

Read instructions.py

4 files +138 -21

instructions.py endpoints
infrastructure prompts
Plan, search, build anything

Agent Ctrl+H claude-4-sonnet



Wrapping up

Lets Summarize Everything.



- Predictive analytics are the tools for better decision making
- Applying Machine Learning in practice is definitely not straightforward
- Scoping is paramount
- Better (and more data) is more important than better models
- Model selection is strategic trade-off
- Consider interpretability when it matters.
- Use LLMs as co-pilots on the different stages of the Predictive analytics lifecycle

Thank you for your attention

Questions?



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<https://github.com/FilTheo>



<https://www.linkedin.com/in/filotas-theodosiou/>



<https://filtheo.github.io/>

